AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2019

Biology Paper II

Time: 2 hours 20 minutes Marks: 40

April/ May 2019 Test for Learning and Learning only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

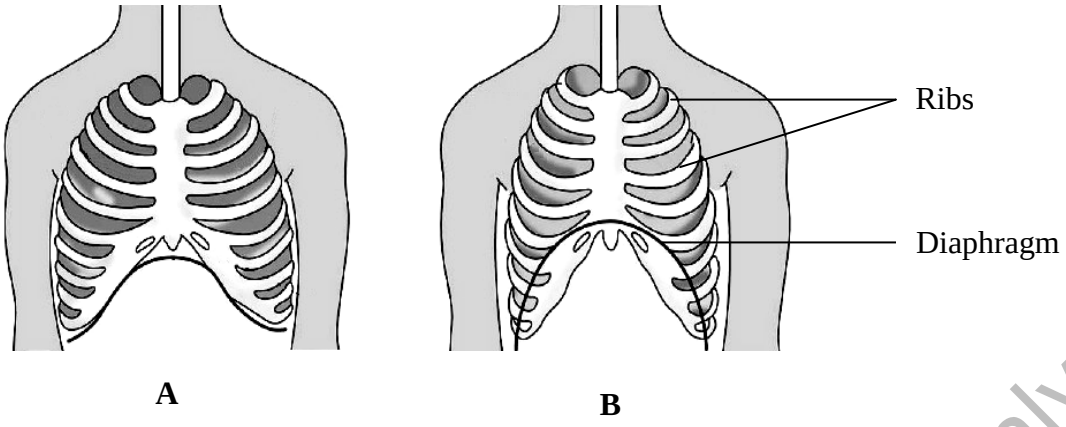
3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. The two steps of breathing are shown in the given diagrams.



Which of the given diagrams, **A** or **B**, shows the process of exhalation? Support your answer with a reason. (2 Marks)

b. Describe the effects of cigarette smoking on the inner walls of trachea of human beings. (2 Marks)

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Q.2. (Total 4 Marks)

a. Mention any TWO ways through which heat is lost from the human body. (2 Marks)

_____

_____

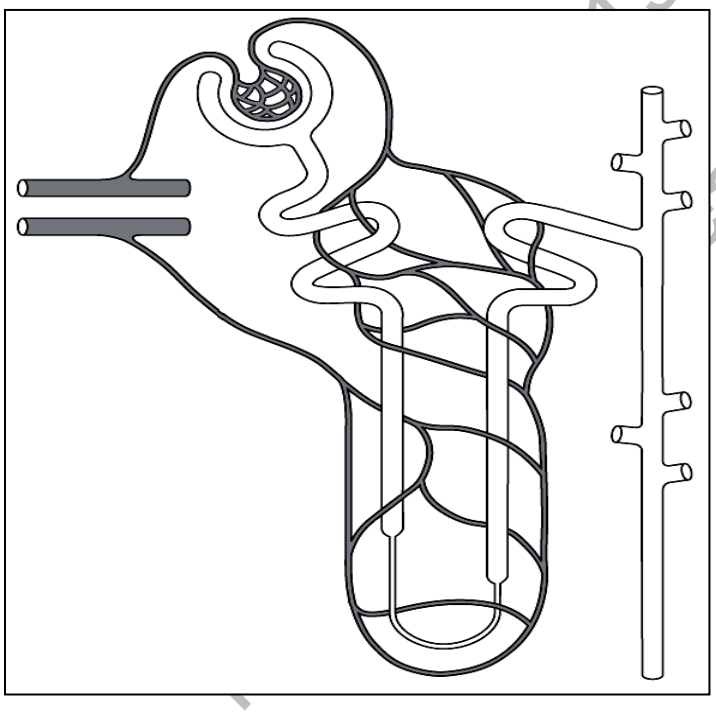
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b. The given diagram shows the structure of a human nephron.

Trace the pathway of blood with the help of arrows (→) in the given diagram. (2 Marks)

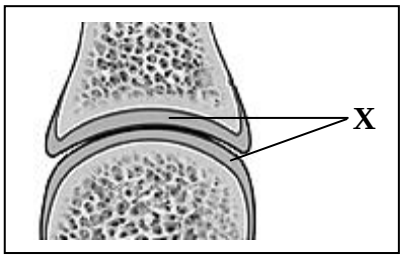
(**Note**: Do NOT show pathway of the filtrate.)



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Q.3. (Total 4 Marks)

The given diagram represents a human hinge joint.



a. Identify structure **X** in the given diagram. (1 Mark)

_____

_____

b. Name the cells that make structure **X**. (1 Mark)

_____

_____

c. Mention the role of structure **X** in the joint. (1 Mark)

_____

_____

d. Name the disease that occurs due to the degeneration of structure **X**. (1 Mark)

_____

_____

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Q.4. (Total 4 Marks)

The given table demonstrates a genetic cross between two flowers.

Parents	Red	×	White	
F1 Generation	Pink			
F1 x F1	Pink	×	Pink	
F2 Generation	Red	Pink	Pink	White

a. Determine the genotype of the pink flowers produced in F1 generation. (1 Mark)

_____

_____

b. Identify the type of dominance in the given cross. (1 Mark)

_____

_____

c. Determine the phenotype and genotype of offspring flowers if two red flowers from F2 generation are crossed. (2 Marks)

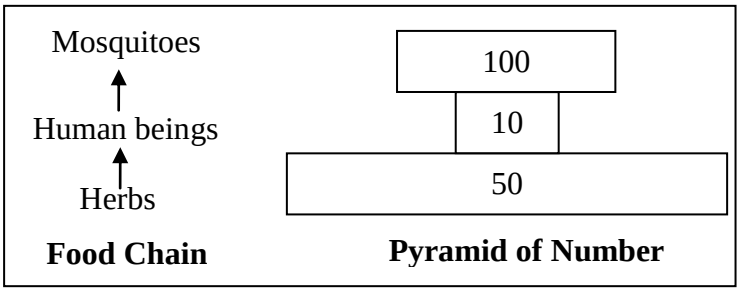
Phenotype: _____

Genotype: _____

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Q.5. (Total 4 Marks)

a. The given diagram represents a food chain and the pyramid of number of the same food chain. (2 Marks)



Interpret the given pyramid of number.

b. Describe the effect of competition on the population of living organisms in an ecosystem. (2 Marks)

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Q.6. (Total 5 Marks)

a. List any THREE steps of genetic engineering. (3 Marks)

b. State the primary usage of the following drugs. (2 Marks)

i. Tetracycline

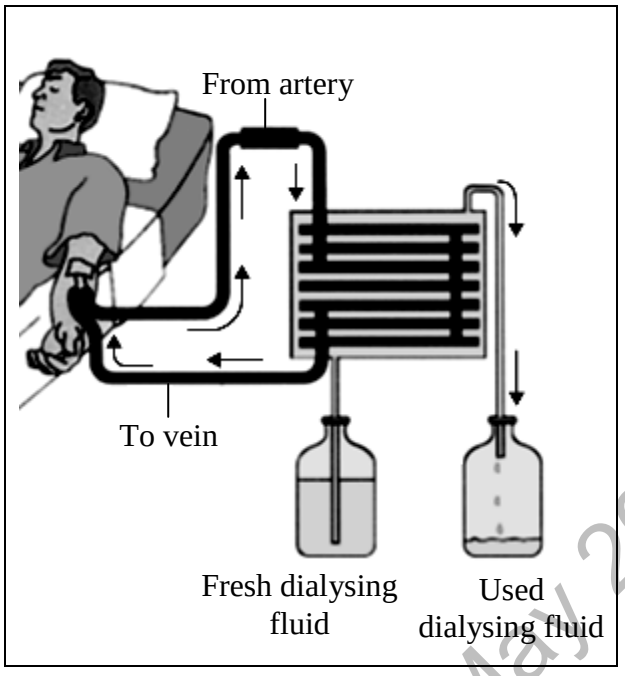
ii. Small pox vaccine

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Q.7. (Total 7 Marks)

EITHER

a.



i. Name the type of dialysis shown in the given diagram. (1 Mark)

ii. Dialysis machine is considered to be an artificial kidney. Describe any THREE features of a dialysis machine on the basis of which it can substitute the functions of different parts of the nephron. (6 Marks)

OR

b.

i. Mention any FOUR physiological changes that would take place in the human body as a result of increased adrenaline level. (4 Marks)

ii. Describe the importance of any THREE physiological changes mentioned in part (i) for exercise or emergency situation. (3 Marks)

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Q.8. (Total 8 Marks)

EITHER

a.

i. Mention any ONE adaptation each of stigma and pollen grain for pollination in an insect pollinated plant. (2 Marks)

ii. Describe the process of sexual reproduction in plants from pollination to fruit formation. (6 Marks)

OR

b.

i. State any TWO sources of variations in sexually reproducing organisms. (2 Marks)

ii. Describe continuous and discontinuous variations. Give an example of human beings in each case to support your answer. (6 Marks)

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END OF PAPER

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